POSTER: Anomaly-Based Misbehaviour Detection in Connected Car Backends

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***Abstract*—As a novel way to protect connected cars, we are developing a Security Information and Event Management System (SIEM) called Security Management of Services in Connected Cars (SeMaCoCa) located in the backend of the connected car. For that we defined a connected car architecture and possible use cases which serve as a basis for the research. Using data from the connected cars and additional information, attacks on individual vehicles or fleets should be recognized. A combination of rule-based-, machine-learning-, deep learning-, real-time-based-, security-algorithms and algorithms for big data are used. Furthermore, we aim for a privacy-friendly solution that does not require the backend operator to have access to cleartext data. The new security system should be able to recognise misbehaviour under the conditions of a permanently growing number and variety of connected cars, upcoming services on the market and related constantly to changing user behavior. The challenge for the security system is, that under these conditions no stable system state exists, that the system can rely on. In this paper, we introduce the architecture of SeMaCoCa, user stories and the idea behind the approach of the system.**

I. Introduction

For high comfort and safety, today's vehicles are equipped with many sensors and control units. However, data that originates in a vehicle is not necessarily confined to it. Each modern car has a so-called Telecommunication Control Unit (TCU), that connects it to a backend via a mobile data connection. This communication can be used for remote control functionality, or for emergency calls in case of accidents. Furthermore, in-car functionality such as entertainment systems or navigation can be updated over the air through the TCU. To be able to offer this kind of services to the user, each car manufacturer typically runs a backend system.

A connected car comes at the same time with many benefits and new security risks. Being able to remotely unlock the car via smart phone could also enable an attacker to do the same thing, should there be security flaws. Similarly, data from the vehicle could be misused if it falls in the wrong hands.

To be able to detect and prevent attacks, we are developing a SIEM called SeMaCoCa, located in the backend of the connected car. With SeMaCoCa, we want to introduce a SIEM that should be able to detect anomalies which can occur on a single car or even fleets. As a detection mechanism, unsupervised machine learning algorithms in combination with real time based algorithms and security algorithms are being adapted and developed. As such, we believe that anomaly-based detection is promising to achieve our goals. As the central system of the connected car is the backend, the entire detection is located here. In the backend, there is direct access to a large number of interesting information. This includes vehicle data (such as odometer values), information from third-party service providers (such as the service status), data from other system participants (like service garages), and internal backend information.

In order to preserve the vehicle users' privacy, we are additionally investigating, parallel to additional concurrent research, the possibility of applying privacy enhancing technologies, that should allow us to achieve the security goals without disclosing any private data even to the system operator.

II. Related Work

Mukkamala et al. [1] are using Support Vector Machines (SVMs) in combination with neural networks to achieve a real time intrusion detection system (IDS) and Li et al. [2] combine SVM, colony algorithms and clustering methods to develop a reliable classifier for intrusion detection in networks. Hui et al. [3] analyse different IDS techniques based on machine learning methods and include methods of neural networking, Bayesian classifiers, data mining and SVM.

Neises[4] introduces the pros and cons of using artificial neural networks, which also relates to deep learning. Also Kuang et al. [5] analyse deep learning algorithms and give an outlook of the direction of deep learning in the future.

III. architecture

The following architecture, an abstract view of a vehicle eco-system, has been defined for the project SeMaCoCa. It is depicted in Figure 1. All use cases and strategies for intrusions and security in the project refer to the given architecture. The architecture defines the subsystems and the interaction of the system participants. Data from all system participants is present in the backend. This makes the backend the central point of the architecture, and thus our SIEM location.

IV. Use Cases

SeMaCoCa is aiming to be able to detect many different attacks and other kind of misbehaviour on and from connected cars. To get a better understanding of the envisioned system, we present a number of use cases in this section.

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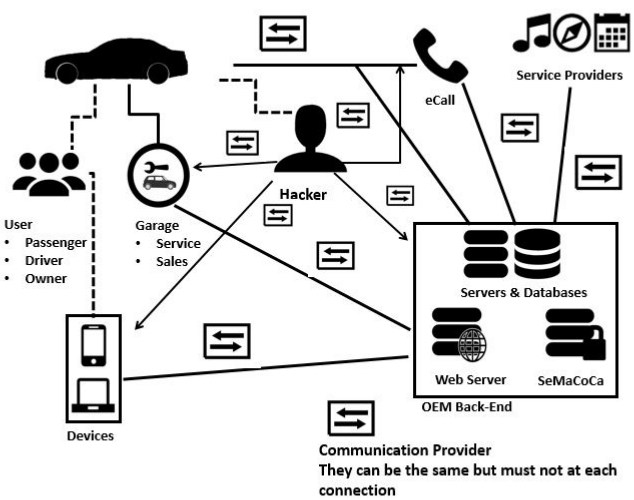

Fig. 1. SeMaCoCa architecture.

a) Special Event or Attack: One of the challenges of SeMaCoCas research is to be able to detect and categorise anomalies from the connected car backend. Anomalies can occur for different reasons. It can be because of an attack on a specific car or on part of a fleet. Further, a harmless but unusual event—such as a soccer game—can cause an anomaly. Another cause for anomalies can be erroneous input data (e.g. from a broken sensor). We need to be able to differentiate what kind of anomaly was detected and assess it. Specifically, whether the abnormal pattern constitutes an attack or other threat.

b) Attack with Stolen Credentials: Another kind of attack—which is difficult to prevent with traditional IT security measures—is the usage of stolen or otherwise obtained credentials. E.g. the attacker might have gotten the password to the connected car account from the owner a long time ago. Now, the attacker uses these credentials to execute remote services which can harm the user. SeMaCoCa should be able to detect this kind of attack by taking into account normal user behaviour.

In all cases, SeMaCoCas should be able to differentiate if benign behaviour is taking place, or an attack. Possibly, system operators might need to be involved when situations are unclear. Also, different warning and security mechanisms can be implemented.

V. SeMaCoCa approach

The aim of SeMaCoCa is to detect anomalies in the connected car backend concerning single cars, or entire fleets. For that a special SeMaCoCa-architecture will be defined. Base for the architecture is the access to all data that is either stored or received by the connected car backend. An algorithm, a combination of unsupervised machine learning algorithms, real time based algorithms and security algorithms, will be developed for SeMaCoCa as well. For that in the first step we have to look into the provided data and filter the data for our kind of misbehaviour detection system; like odometer values, fuel levels, door lock status, positions, etc. After this step we have to define which additional data can improve the detection results and has to be added to SeMaCoCas dataset. This could include i.e. weather data or road conditions. Another requirement for SeMaCoCa is to stay up to date, as the behaviour of the users and the data in the backend are continuously changing. Adapting to new services, is also a requirement of SeMaCoCa, for that each step in the architecture has to be processed cyclical. Further, the system should ideally be able to recognise whether a special event (such as a soccer game which can cause anomalies), is tacking place and is causing a big amount of service activations at the same time and place. Such events should not be categorised as an attack. This can be done by observing different event calendars on the internet. Because the response time of SeMaCoCa—e.g. in case of theft—has to be fast, the entire system has to be able to operate in real-time. We aim for a combination of a rule-based and self-learning approach. Anomalies and attacks should be detected by machine learning algorithms, but expert knowledge and appropriate rules should be able to increase the accuracy of the system. Another important aspect of building a system that monitors an entire fleet of vehicles is the privacy of the car users. Thus, in concurrent research, we investigate how privacy enhancing technologies can be applied, so that the privacy of the users is preserved and system operators do not learn too much private information.

VI. Conclusion

We introduced a connected car architecture which is used as a base for a SIEM system called SeMaCoCa. Also the main use cases which the system should be able to handle were presented. As SeMaCoCa is envisioned as an anomaly-based detection system, we currently investigate different unsupervised and semi-supervised learning methods for their suitability to the system. In the next steps we will build a simulation based on the use cases.

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